54. Capstone: Broken Box Challenge

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Phase: Phase 12 — Capstones
Estimated time: 2–4 days
Slug: `54-capstone-troubleshoot`

Learning objectives

- Debug a deliberately broken VM
- Capture timeline and root cause
- Fix with minimal change
- Write postmortem

Prerequisites

Complete prior numbered lessons. Prefer a real Linux VM/container for Linux-specific topics.

Mission

Someone (you, earlier) breaks a snapshot on purpose: full disk, bad fstab, broken DNS, failed unit, wrong permissions, expired cert — pick 3+.

Restore service with a written postmortem:

- Impact
- Timeline
- Root cause
- Fix
- Prevention

Rubric

- Structured troubleshooting
- Evidence preserved
- Minimal blast radius
- Clear prevention items

Common mistakes

- Recreating VM instead of debugging
- Changing everything at once
- No postmortem

Exercises

Work in `exercises/54-capstone-troubleshoot/`. Run:

```
./test.sh
```

- **A (Drill):** make `./test.sh` pass
- **B (Apply):** note one real command you ran on a Linux host related to this lesson
- **C (Challenge):** rewrite your solution from memory the next day

Do not open `solutions/54-capstone-troubleshoot/` until you have an honest failing attempt.

Self-check

1. What was root cause?
2. What evidence?
3. What prevention?

Further reading

- https://sre.google/sre-book/postmortem-culture/